Individual Homework 05 – Norhelia Echevarria

1. PGD Evasion Audit (LR vs. GBT, ε ∈ {0.25, 0.5, 1.0, 2.0})

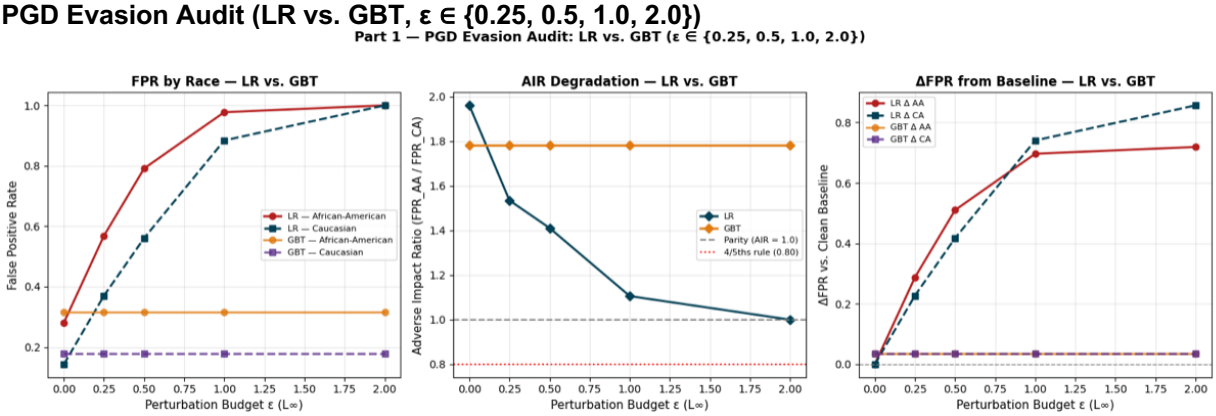

Figure 1. FPR by Race, AIR Degradation and ΔFPR from Baseline - LR vs. GBT

From Figure 1, the main takeaway is that LR and GBT respond very differently to the PGD attack. First, looking at the FPR by race, LR clearly breaks under the attack. As ε increases, the FPR rises very quickly for both African-American and Caucasian groups, reaching almost 1.0. This means the model ends up flagging almost everyone as positive, which makes it unusable in practice. In contrast, GBT remains almost unchanged, its FPR stays flat across all ε values, showing strong robustness to this type of attack. Second, the AIR panel shows an interesting effect. For LR, AIR decreases toward 1.0 as ε increases. At first glance, this might look like an improvement in fairness, but it is actually misleading. The reason is that both groups are being harmed more equally, not that the model is becoming fair. This is a clear example of what we discussed in class as fairness through harm. Meanwhile, GBT's AIR stays constant around 1.78, meaning it is stable but already unfair from the beginning. Finally, the ΔFPR panel confirms this pattern. The increase in false positives is larger for Caucasian than for African-American at higher ε values in LR, which explains why the gap closes. Again, this is not a real fairness improvement but a deterioration affecting both groups. Overall, this figure shows a clear tradeoff. LR is highly vulnerable to adversarial attacks, while GBT is much more robust. However, GBT's baseline fairness issue remains unresolved. So neither model is reliable as-is, especially in a high-stakes setting like criminal justice, where both robustness and fairness are critical.

2. Label-Flip Poisoning: African – American vs. Caucasian

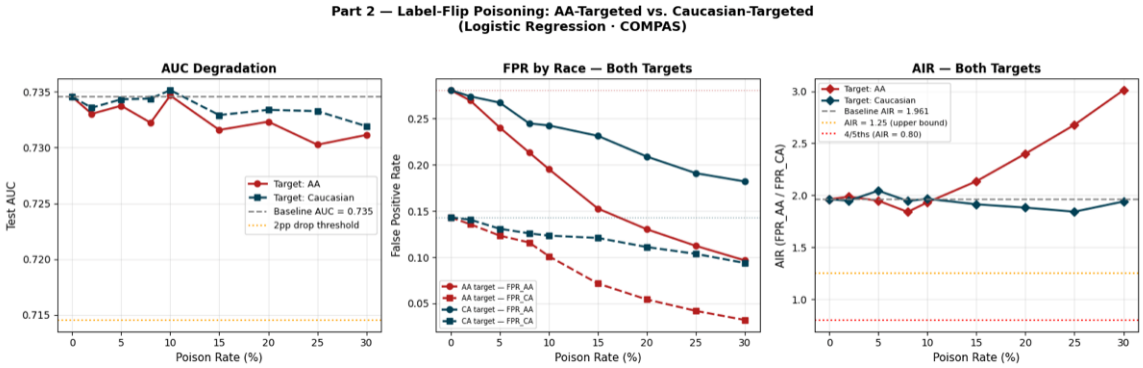

Figure 2. AUC Degradation, FPR by Race, and AIR - Both Target Groups

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Part 2(b) — Stealth Zone Analysis
  Stealth = |delta_AUC| ≤ 0.02  AND  AIR outside [0.80, 1.25]
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  ★ AA-Targeted — Stealth zone STARTS at poison_rate = 0%
    n_flipped = 0
    AUC       = 0.735  (Δ = +0.000)
    AIR       = 1.961  (Δ = +0.000)
    Stealth zone spans rates: 0% — 30%

  ★ CA-Targeted — Stealth zone STARTS at poison_rate = 0%
    n_flipped = 0
    AUC       = 0.735  (Δ = +0.000)
    AIR       = 1.961  (Δ = +0.000)
    Stealth zone spans rates: 0% — 30%
```

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Part 2(c) — Would PSI-Based Drift Detection Catch These Attacks?
  PSI threshold: < 0.10 = stable (no alert)
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--- PSI on INPUT FEATURES under poisoning ---
(Label-flip does NOT change feature values → PSI must be 0)

  AA-Targeted: checking feature PSI at max poison rate (30%)...
    Max feature PSI across all features = 0.000000  → NO ALERT (PSI < 0.10)
  CA-Targeted: checking feature PSI at max poison rate (30%)...
    Max feature PSI across all features = 0.000000  → NO ALERT (PSI < 0.10)

--- PSI on MODEL OUTPUT SCORES under poisoning ---
(Score-PSI is a more sensitive lagging indicator)

  AA-Targeted:
    rate=0.00  Score PSI=0.0000  → no alert
    rate=0.02  Score PSI=0.0027  → no alert
    rate=0.05  Score PSI=0.0163  → no alert
    rate=0.08  Score PSI=0.0287  → no alert
    rate=0.10  Score PSI=0.0552  → no alert
    rate=0.15  Score PSI=0.0983  → no alert
    rate=0.20  Score PSI=0.1740  → ALERT ⚠
    rate=0.25  Score PSI=0.2320  → ALERT ⚠
    rate=0.30  Score PSI=0.3318  → ALERT ⚠

  CA-Targeted:
    rate=0.00  Score PSI=0.0000  → no alert
    rate=0.02  Score PSI=0.0017  → no alert
    rate=0.05  Score PSI=0.0054  → no alert
    rate=0.08  Score PSI=0.0075  → no alert
    rate=0.10  Score PSI=0.0151  → no alert
    rate=0.15  Score PSI=0.0202  → no alert
    rate=0.20  Score PSI=0.0479  → no alert
    rate=0.25  Score PSI=0.0554  → no alert
    rate=0.30  Score PSI=0.0783  → no alert
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Figure 3. Stealth Zone Analysis (left) · PSI Detection Output (right)

Looking at Figure 2, both attacks are very stealthy in terms of performance but still create strong fairness problems. AUC stays almost unchanged across all poison rates, always within the 2pp threshold, so no performance-based alert would be triggered. The FPR panel shows how the attacks work. In the AA-targeted attack, FPR decreases for both groups, but faster for African-American. In the CA-targeted attack,

FPR for African-American stays stable while FPR for Caucasian decreases, increasing the gap between groups. The AIR panel confirms this: the AA-targeted attack pushes AIR above 3.0 (worsening disparity), while the CA-targeted attack keeps AIR around 1.9–2.0, still outside the acceptable range. Both attacks operate entirely within the stealth zone (0%–30%), since baseline AIR is already outside the fairness range and AUC barely changes. This means the model could be under attack without any detection if we only monitor performance. PSI also fails to detect these attacks. Feature PSI is 0 because only labels are modified. Score PSI only detects the AA-targeted attack at high poison rates (~20%+), and never detects the CA-targeted attack.

Overall, the key takeaway is that these attacks can harm fairness while remaining invisible to AUC and PSI, so fairness-specific monitoring (like AIR and FPR by group) is necessary.

3. Membership Inference Depth: LR vs. GBT + L2 Regularization Sweep

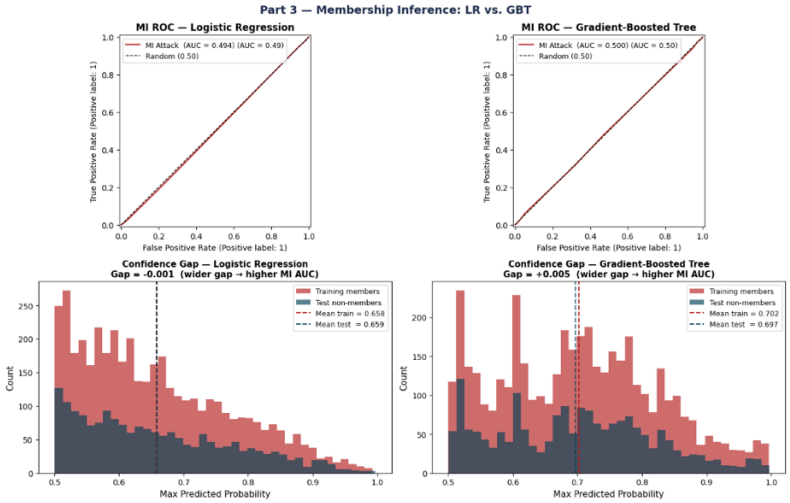

Figure 4. MI ROC Curves and Confidence-Gap Histograms — LR vs. GBT

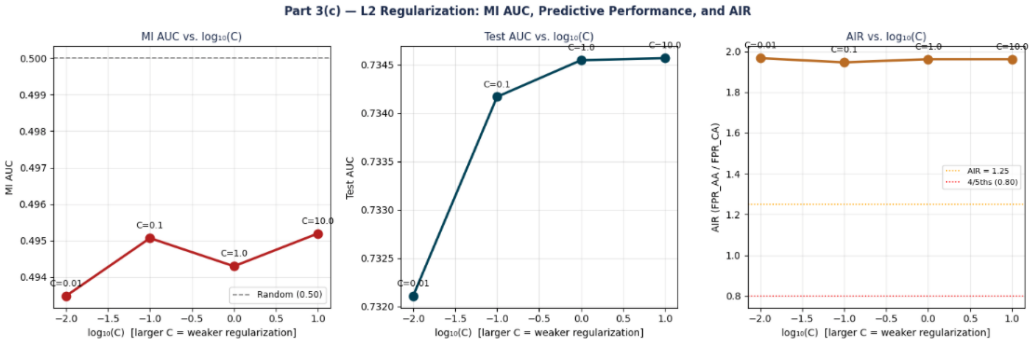

Figure 5. L2 Regularization Sweep: MI AUC, Test AUC, and AIR vs. log10(C)

(a) Confidence-Gap Histograms: Both models show almost complete overlap between training and test distributions, which explains why MI AUC is basically random (LR = 0.494, GBT = 0.500). The confidence gaps are very small (LR = −0.001, GBT = +0.005), so there is no clear signal that an attacker could exploit.

(b) Generalization Gap vs. MI AUC: The results support the theory directionally: GBT has a larger generalization gap (+0.080) and slightly higher MI AUC than LR. However, both MI AUC values are essentially random, so the practical privacy risk is negligible.

(c) L2 Regularization and Tradeoff: Changing C has almost no effect on MI AUC (stays around 0.493–0.495), so there is no privacy improvement. Strong regularization (C = 0.01) slightly reduces test AUC (0.735 → 0.732), meaning worse performance with no benefit. AIR stays constant (~1.96), so fairness does not improve.

Overall, C = 1.0 is the best balance since regularization does not improve privacy or fairness but can hurt performance.

4. Reflection: Highest-Risk Finding and Mitigations

The highest-risk finding is the label-flip poisoning attack. It is dangerous because it is realistic (anyone with data access could do it), completely invisible to standard monitoring (PSI = 0 and AUC barely changes), and it can affect fairness in different ways depending on the target group. In contrast, the membership inference results show no real privacy risk since MI AUC is basically random.

For proactive mitigation, using data provenance (hashing the dataset with SHA-256) can fully prevent this attack. Any label change would modify the hash, so the poisoned data would not enter training. This does not introduce bias, but it could slow down legitimate data corrections, which can be managed with review protocols.

For reactive mitigation, replacing PSI with fairness monitoring (tracking AIR and FPR by race) is more effective. This would have detected the issue immediately since the baseline AIR is already outside the acceptable range. Score-PSI can help partially, but it only detects some cases and too late.

Overall, the key takeaway is that PSI-based monitoring is not enough, fairness monitoring is essential to detect these types of attacks.